



Content Management Systems

CS571: Building User Interfaces

Adapted from slides provided by...

CS272: Introduction to Web Development

Learning Objectives

After this lecture, students should be able to...

1. Define what a Content Management System (CMS) is
2. Compose a basic website with a CMS (WordPress), including the use of...
 1. Pages
 2. Posts
 3. Patterns
 4. Navigation
 5. Themes
 6. Plugins
3. Understand the pros and cons of hosting their own WordPress site.

What is a Content Management System (CMS)?

- A type of server.
- A content management system (CMS) is software that helps users create, manage, store, and modify their digital content. (IBM)
- A CMS, short for content management system, is a software application that allows users to build and manage a website without having to code it from scratch, or know how to code at all. (HubSpot)

WordPress

- A type of Content Management System.
- “The open-source publishing platform of choice for millions of websites worldwide—from creators and small businesses to enterprises.” – wordpress.org
- Approximately 20-40% of websites use WordPress!
- Not to be confused with *wordpress.com*!
 - *wordpress.org* – the open-source WordPress software
 - *wordpress.com* – a paid host for WordPress
 - WPEngine, DreamHost, Hostinger, UW-Madison CS Dept, among others...

WordPress Essentials

- The *theme* dictates how *pages* and *posts* are displayed.
- Both **pages** and **posts** hold content; this could be...
 - Headings
 - Text
 - Images
 - ... and much more!
- **Pages** are intended for static content that doesn't change frequently.
- **Posts** are intended for blog entries or articles that are time-sensitive.
 - **Posts** can also be *tagged* and *categorized*.



Let's Play **Page Vs. Post**

I give you a title, you tell me if it's a page or post!

Page or Post?

About Us



Page or Post?

Badger Bytes: August Edition



Page or Post?

Contact Us



Page or Post?

Privacy Policy



Page or Post?

My Trip to Jamaica!



Page or Post?

FAQ



Page or Post?

College Dorm Room Cleaning: Tips & Tricks

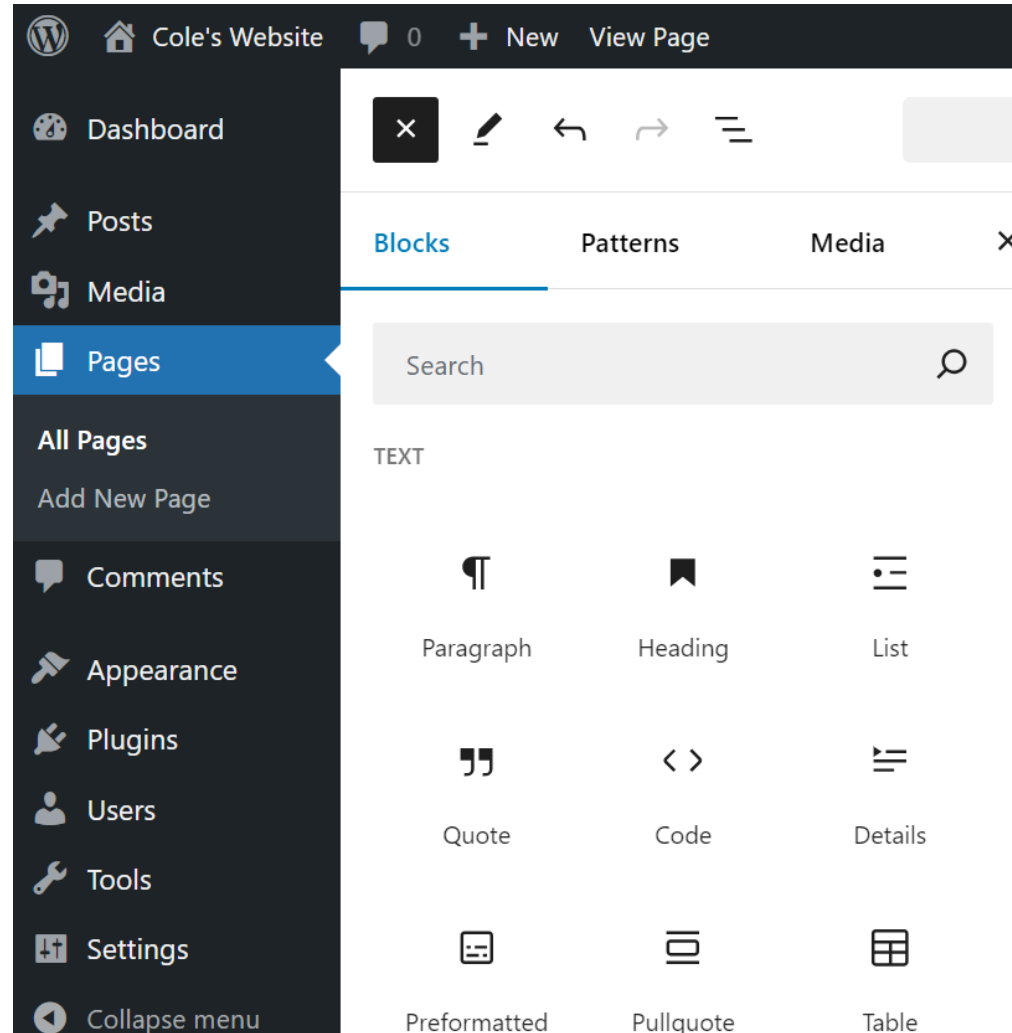




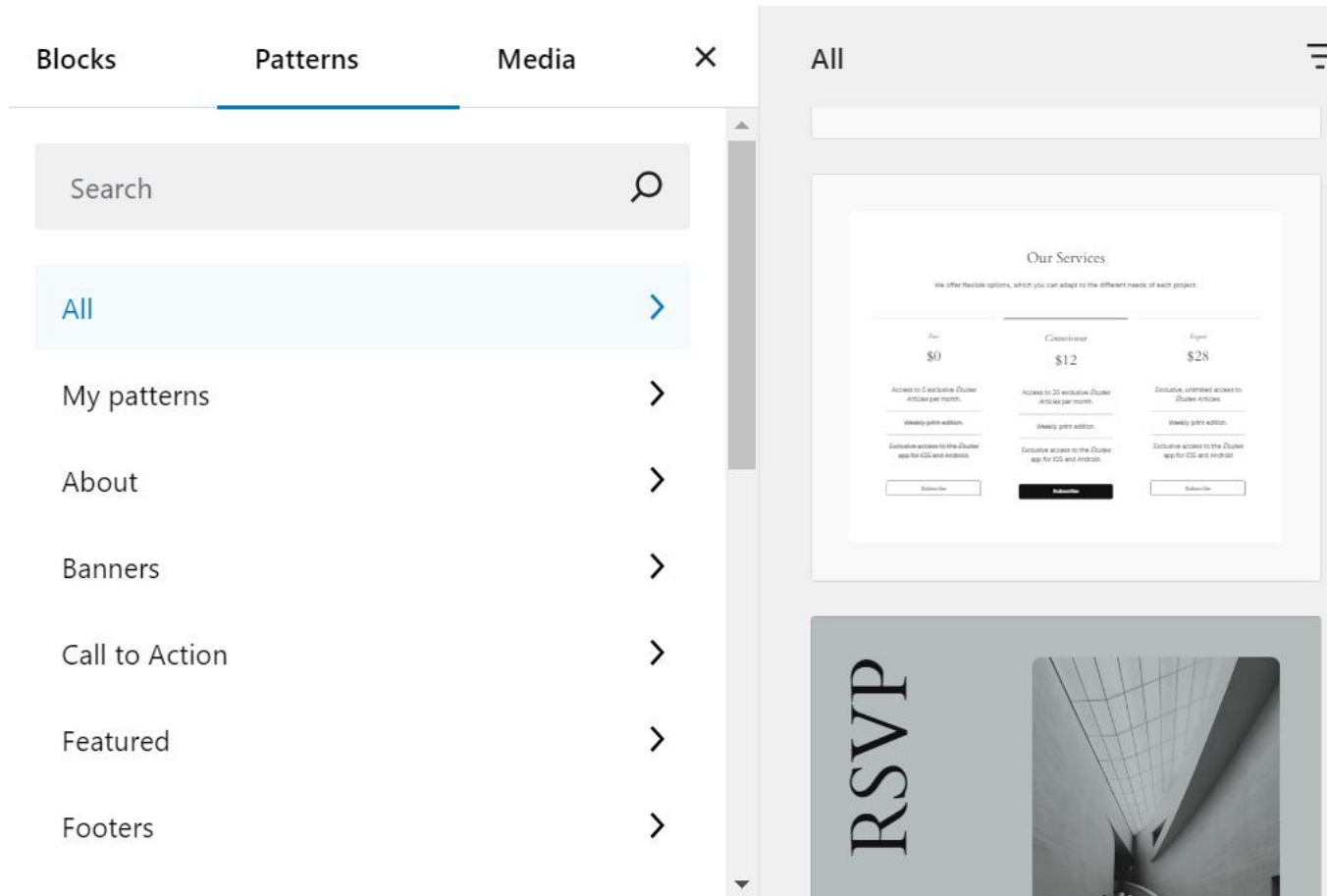
Building Pages & Posts

How do we add content to our pages and posts?

The Gutenberg Editor



Patterns



Generally accepted formats for combining blocks into a typical use case.

e.g. project description, pricing, calls to action, etc.

Your theme provides some patterns, **but you can also create your own!**

Using WordPress Media

- Any media type is important, but certain types may have additional attributes.
- Every image should have appropriate alt text.
 - Captions and descriptions are also helpful, but not totally necessary!
 - Captions can be included in the Block Editor
- Be careful when you upload something! It becomes *immediately* available to *everyone*, so long as they have a URL.
 - Pretend that *anything* you upload is public!



Let's dive into WordPress!

Replace with your NETID!



<https://cs272-wordpress.cs.wisc.edu/su25/ctnelson2-site/>

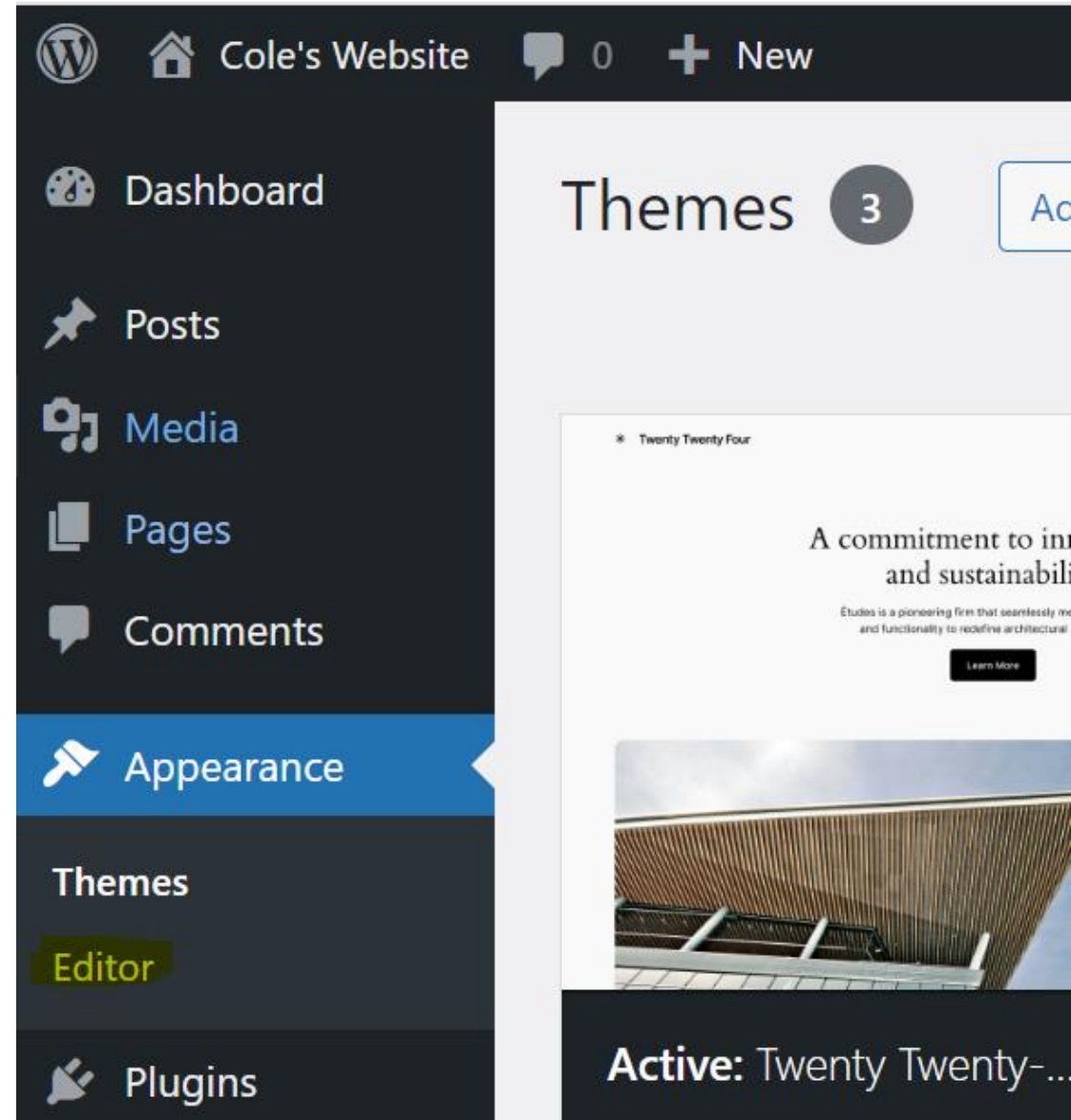
⚠ You MUST be connected [to the UW-Madison VPN](#).

Patterns

Patterns can be managed via the full-site-editor.

This is a new feature of some WordPress themes!

Be careful what you change, it could affect your entire site!



Patterns

Patterns can be **synced** or **not synced**.

A pattern that is **synced** will look the same everywhere that it is used!

A pattern that is **not synced** can be changed to meet the needs of a specific part of the webpages that it is being used.

All patterns

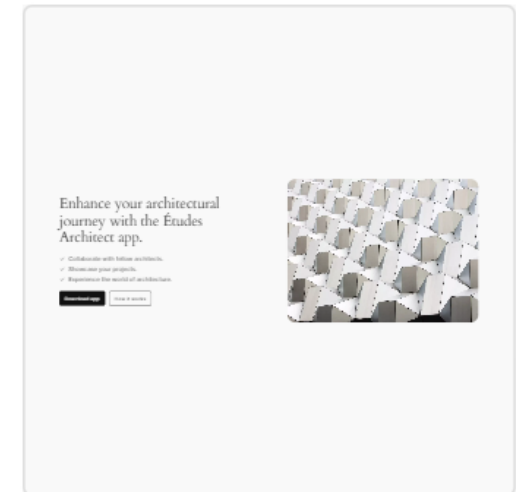
A list of all patterns from all sources.



☐ Copyright



Synced



☐ Call to action wi...



Not synced



A **template part** is a *synced pattern* used in specific parts of a theme (e.g. header, footer)



Unsynced Patterns, Synced Patterns, and Template Parts

	Template Part	Synced Patterns (Reusable Block)	Patterns
Type	Site Structure	User Content	User Content
Syncing Ability	Synced	Synced	Un-synced
Example	Footer	Business hours	Call to action

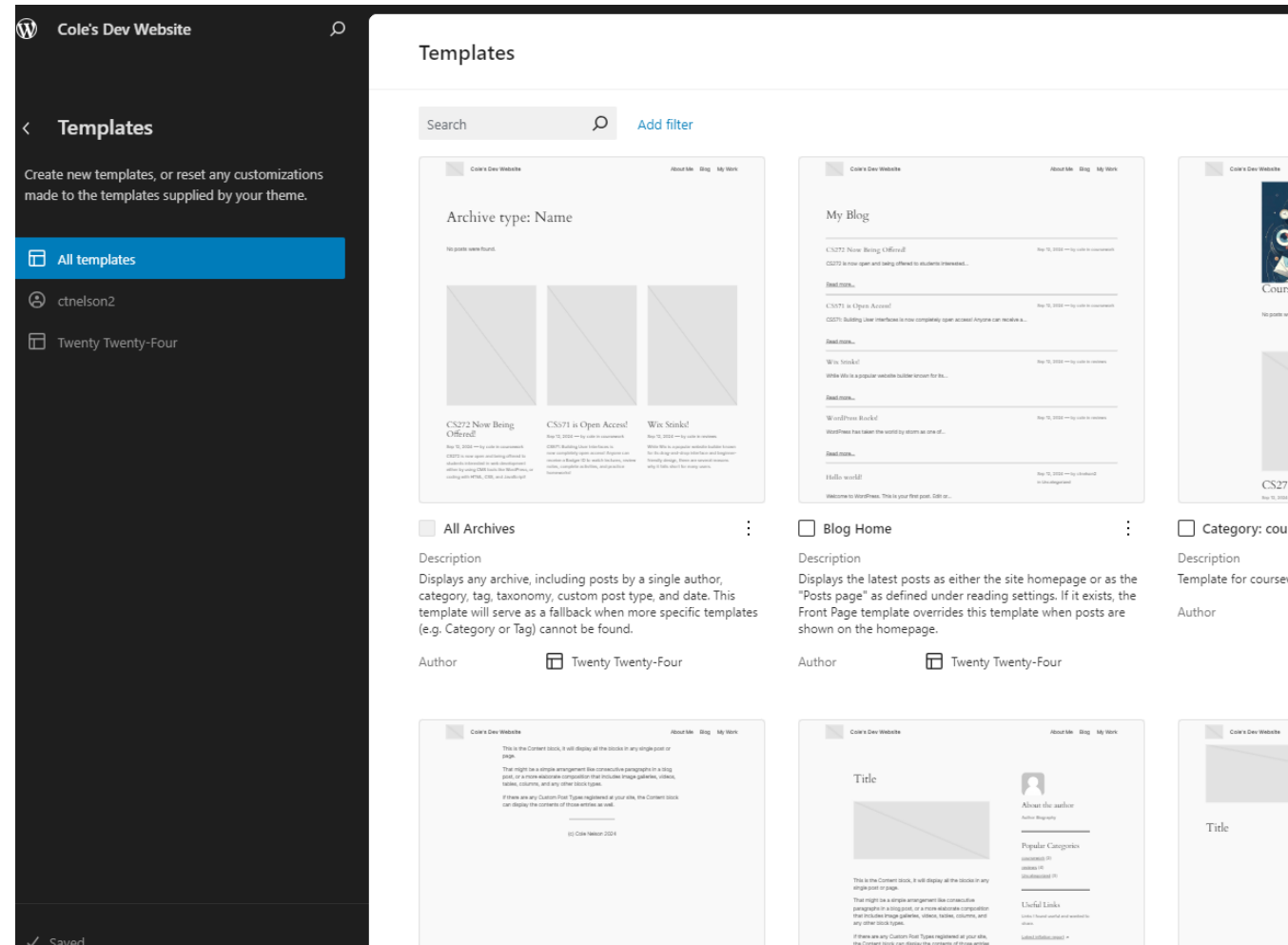
[View the full article here!](#)

Templates

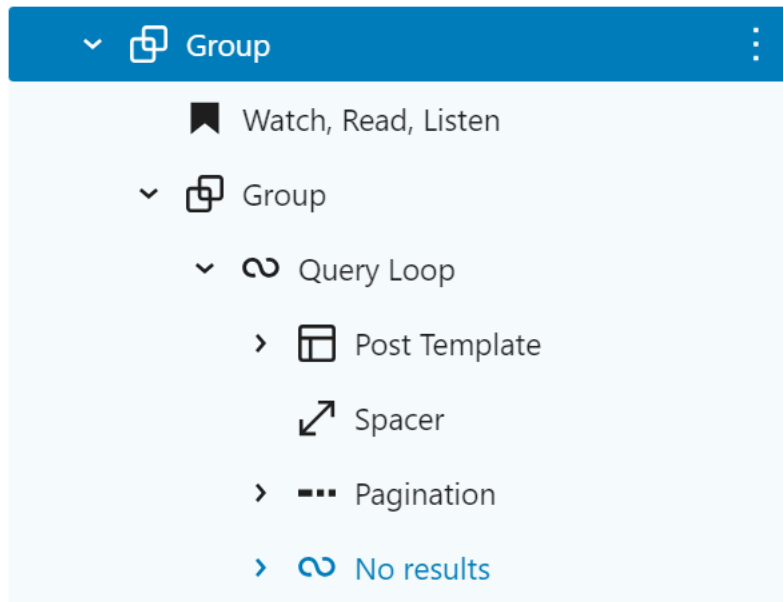
Define specific representations for the website, as dictated by the theme.

Often make use of many “template parts”.

e.g. **Blog Home** is a template that defines your homepage (for now!)



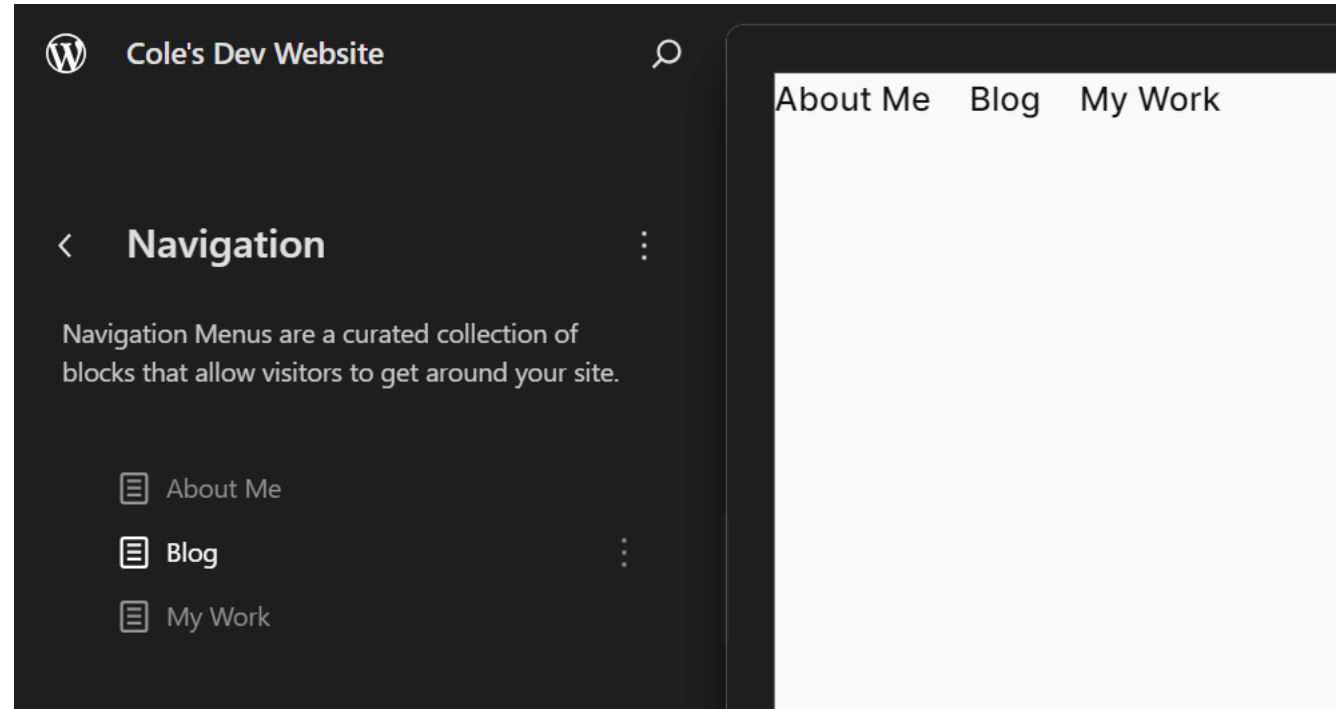
Templates



Template often make use of blocks we haven't seen before!

These usually consist of dynamic content, that changes based on the page that is being displayed.

Navigation



Editing the Navigation will cause it to de-sync, meaning you'll have to manually add pages in the future. **That's okay!**

Your turn!

Create and/or modify **patterns**, **template parts**, **templates**, and the **navigation** of your site!

Themes

- Dictate the overall layout and appearance of your website, including...
 - Default fonts, colors, and other CSS styles
 - Available templates and template parts
 - Navigation options
- WordPress team themes vs custom themes
 - Custom themes can be free, freemium, or premium!
- Block themes (Full-Site Editing) vs Classic Themes
 - e.g. Twenty-Twenty-Four vs. Twenty-Twenty
 - Classic themes do not have a full-site editor!

Plugins

- Can be free, freemium, or premium!
- Add additional features to WordPress such as...
 - Search-Engine Optimization (SEO), e.g. Yoast
 - Auditing and tracking, e.g. Simple History
 - Security features, e.g. WPS Hide Login
 - Forums, e.g. bbPress
 - Forms, e.g. Contact Form 7
 - Much, much more!
- Vary *greatly* in size and complexity.

Not an endorsement of any particular plugin.

Plugins and Themes

- Plugins and themes from the official wordpress.org marketplace are “trusted”.
 - This does not mean they are “endorsed”!
 - There *may* still be security vulnerabilities!
- What impact might custom plugins and themes have on security and maintainability of your site?

Self-Hosted vs Hosted WordPress Sites

Self-Hosted WordPress

e.g. running on your computer

- High technical skill required
- Only available when your computer is on
- Visitors connect to *you*
- No restrictions on WordPress content
- Free*

Hosted WordPress

e.g. running [on a dedicated host](#)

- Low technical skill required
- Available 24/7
- Visitors connect to *your host*
- Often has restrictions on WordPress content
- Paid

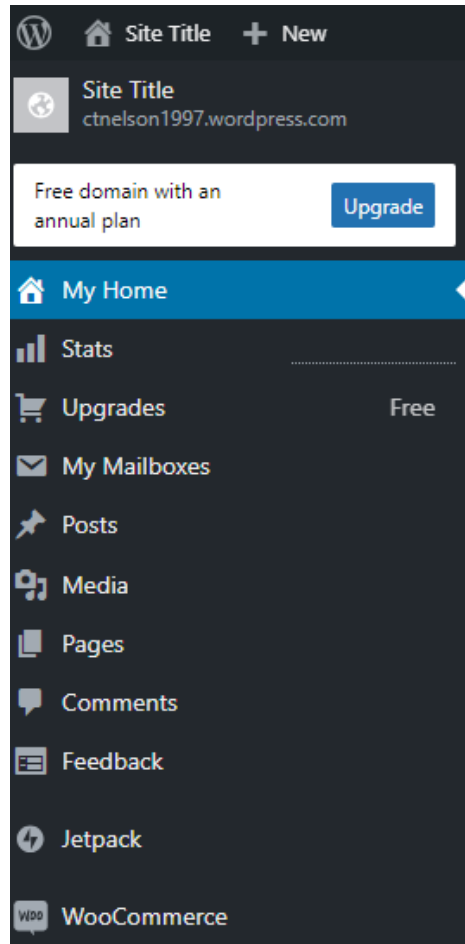
If you are interested in self-hosting a WordPress site, [see XAMPP](#) or [Docker](#)

Hosted WordPress Sites

Free	Personal	Popular Premium	Best for devs Business	WooCommerce Commerce	VIP Enterprise
Get a taste of the world's most popular CMS & blogging software. \$0 No expiration date Start with Free	Create your home on the web with a custom domain name. \$4 per month, \$48 billed annually, excl. taxes Get Personal	Build a unique website with powerful design tools. \$8 per month, \$96 billed annually, excl. taxes Get Premium	Unlock the power of WordPress with the managed hosting platform built by WordPress experts. \$25 per month, \$300 billed annually, excl. taxes Get Business	Build an online store with powerful, integrated premium extensions. \$45 per month, \$540 billed annually, excl. taxes Get Commerce	Deliver an unmatched performance with the highest security standards on our enterprise content platform. Starts at \$25,000.00 yearly Learn more
Beautiful themes and patterns Unlimited pages Unlimited users Time machine for post edits Built-in newsletters & RSS Brute-force protection Smart redirects Online forever 10% transaction fee for payments (+ standard processing fee)	Everything in Free, plus: Free domain for one year Access to dozens of premium themes Ad-free experience Extremely fast DNS with SSL 8% transaction fee for payments (+ standard processing fee)	Everything in Personal, plus: Free domain for one year Fast support from our expert team Access to all premium themes Earn with WordAds Customize fonts and colors 4% transaction fee for payments (+ standard processing fee)  Upload 4K videos with VideoPress	Everything in Premium, plus: Free domain for one year Priority 24/7 support from our expert team Install plugins & themes Unrestricted bandwidth No limitations on site visitors Global edge caching High-burst capacity Web application firewall (WAF) Global CDN	Everything in Business, plus: Free domain for one year Optimized WooCommerce hosting Premium store themes Powerful store design tools Unlimited products Display products by brand Product add-ons Assembled products and kits Min/max order quantities Back-in-stock notifications	High performance platform, with: Multifaceted security Generative AI                

From wordpress.com

Hosted WordPress Sites

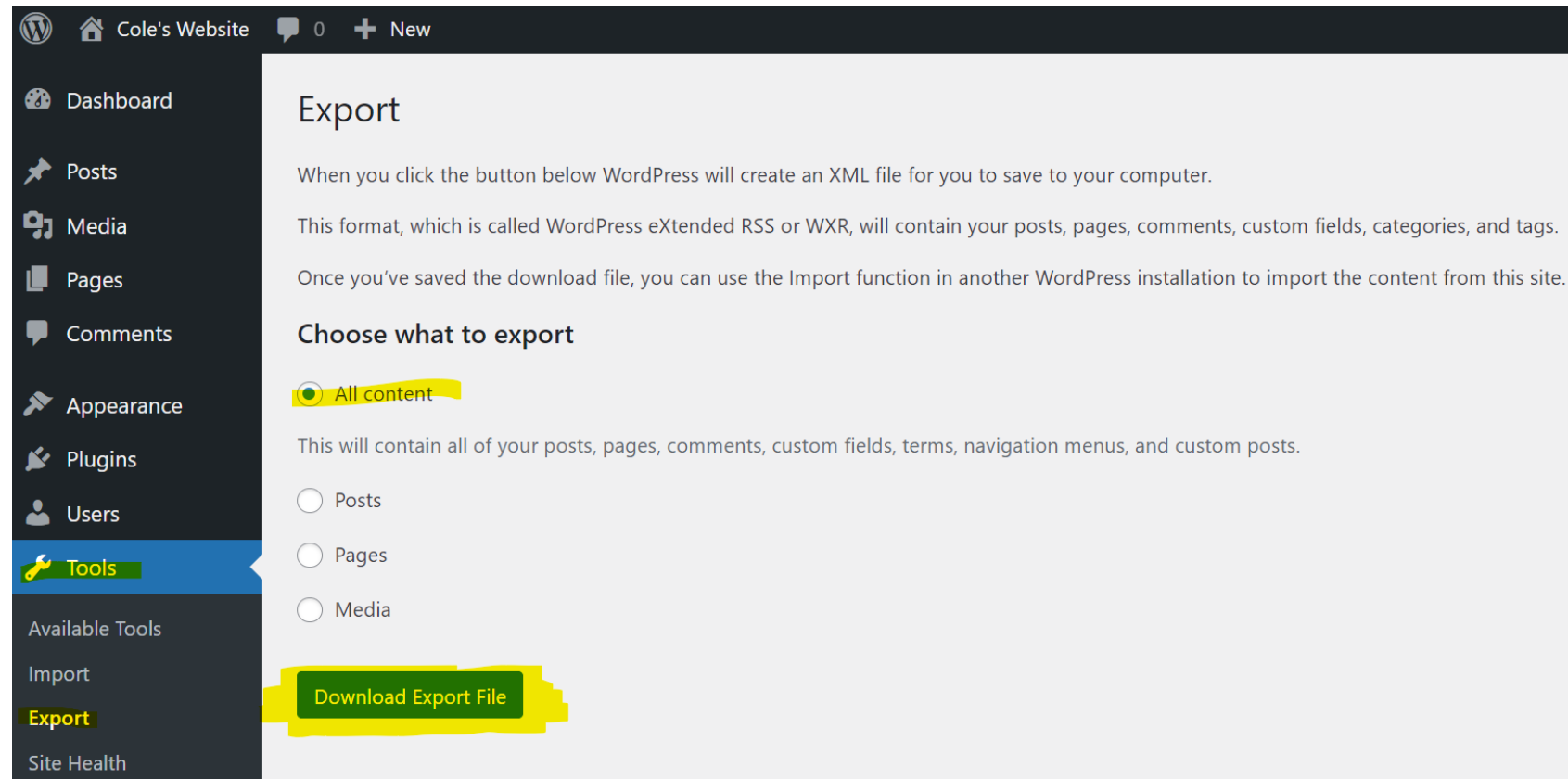


Because they host your WordPress site, they get to add/remove WordPress content on your behalf.

What you see will be a little bit different from what we've done with wordpress.org

From wordpress.com

Exporting Your Site



If you use cs.wisc.edu, websites will be removed after the term! **Export.**

